

Homework5 CS5800

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1 Exercise

(a)

If we select three random elements p_1, p_2, p_3 we first sort them (in ascending order) which would take $O(1)$. Assume $p_1 < p_2 < p_3$. Then we partition the array using a single pass and partitioning it into 4 groups: elements less than p_1 , elements between p_1 and p_2 , elements between p_2 and p_3 , and elements higher than p_3 . This would take $O(n)$, since each element is compared against the three pivots once. Finally, we determine the ranks. Rank of p_1 is the number of elements less than $p_1 + 1$. Rank of p_2 is the rank of p_1 + number of elements between p_1 and $p_2 + 1$. Rank of p_3 is the rank of p_2 + number of elements between p_2 and $p_3 + 1$. These take $O(1)$. This means that in total, the complexity is less than $3dn$.

(b)

The number of elements outside of $[\frac{n}{4}, \frac{3n}{4}]$ is $n - (\frac{3n}{4} - \frac{n}{4}) = n - \frac{n}{2} = \frac{n}{2}$. Therefore, the probability that a single randomly chosen element has rank outside of $[\frac{n}{4}, \frac{3n}{4}]$ is $\frac{n/2}{n} = \frac{1}{2}$. Since the ranks of the three elements are independent, the probability that all three have ranks outside of $[\frac{n}{4}, \frac{3n}{4}]$ is $(\frac{1}{2})^3 = \frac{1}{8}$.

(c)

The probability that at least one of the three randomly chosen elements has a rank in $[\frac{n}{4}, \frac{3n}{4}]$: $1 - P(\text{all outside}) = 1 - \frac{1}{8} = \frac{7}{8}$. So the expected number of trials until then is $\frac{1}{7/8} = \frac{8}{7}$. So $T(n) \leq \frac{8}{7}O(n) + T(\frac{3n}{4})$. $T(n) \leq \frac{8}{7}n + \frac{8}{7}\frac{3n}{4} + \frac{8}{7}(\frac{3}{4})^2n + \dots$ Geometric series with $r = \frac{3}{4}$. So we get $T(n) \leq \frac{8}{7}n4$, which is $T(n) \leq 4.57n$ (so worse leading constant than standard algorithm).